

學號：_____ 姓名：_____

作答後交回

題號	一 選擇題	二	三	四	五	六	七	八	總分
得分	40	9	12	6	5	8	10	10	

一 選擇題(每題 4 分)

1. a The data in ? is erased if the computer is powered down.
(a) **RAM** (b) ROM (c) HARD-DISK (d) CD-ROM (e) DVD
2. a ? is a memory type with capacitors that need to be refreshed periodically.
(a) **DRAM** (b) ERAM (c) SRAM (d) ROM (e) EROM
3. d In the ? method for synchronizing the operation of the CPU with an I/O device, the I/O device informs the CPU when it is ready for data transfer.
(a) serial I/O (b) parallel I/O (c) programmed I/O (d) **interrupt-driven I/O** (e) DMA
4. c In the ? method for synchronizing the operation of the CPU with an I/O device, the CPU is idle until the I/O operation is finished.
(a) serial I/O (b) parallel I/O (c) **programmed I/O** (d) interrupt-driven I/O (e) DMA
5. c In the TCP/IP protocol suite, the _____ layer is responsible for route determination.
(a) physical (b) data link (c) **network** (d) transport (e) application
6. b In a computer network, various layers work together, each performing specific roles to ensure data communication. In TCP/IP protocol suite, the primary protocol at network layer is ? .
(a) TCP (b) **IP** (c) UDP (d) SCTP (e) NLC
7. b In the TCP/IP protocol suite, the _____ layer is responsible for handling frames between adjacent nodes.
(a) physical (b) **data link** (c) network (d) transport (e) application
8. b An IPv4 address is ? bits in length.
(a) 16 (b) **32** (c) 64 (d) 128 (e) 196
9. d An IPv6 address is ? bits in length.
(a) 16 (b) 32 (c) 64 (d) **128** (e) 196
10. d ? is a protocol for accessing and transferring documents on the WWW.
(a) FTP (b) SMTP (c) TELNET (d) **HTTP** (e) DNS

填充與問答題

二 (9 分) The CPU uses repeating machine cycles to execute instructions in the program, one by one, from beginning to end. What are the three phases in a simplified machine cycle?

(a) Fetch (b) Decode (c) Execute

三 (12 分) An imaginary computer has 8 data registers (R0 to R7), 512 words in memory, and 16 different

instructions (add, substrate, and so on).

(a) What is the minimum size of an add instruction in bits if a typical instruction uses the format:

add R1 R2 M1 (這個指令最少要幾個bits?) Ans: 4+3+3+9 = 19 bits

(b) What is the size of the instruction register? Ans: 19 bits

(c) What is the size of the program counter? Ans: 9 bits

(d) If the computer uses the same size of word for data and instructions. What is the size of the data bus in the computer? 19 bits

(e) What is the size of the address bus in the computer? Ans: 9 bits

(f) What is the minimum size of the control bus in the computer? Ans: 4 bits

四 (6 分) A network must be able to meet a number of criteria. Identify the three most important ones.

(a) Performance (b) Reliability (c) Security

五 (8 分) The term physical topology refers to the way in which a network is laid out physically. Two or more devices connect to a link; one or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and devices to one another. What are four basic topologies of networks?

(a) Mesh (b) Star (c) Bus (d) Ring

六 (5 分) TCP/IP is a protocol suite used in the Internet today. It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality. Name the layers of the TCP/IP protocol suite. 列出 TCP/IP 通訊協定 5 層

Ans: L1: physical layer L2: data link layer L3: network layer
L4: transport layer L5: application layer

以下兩題寫出程式的機器碼，每個指令用 4 個 16 進位數字(digit)表示，如 1040

七 (10 分) Using the instruction set of the simple computer as described in the class, write the code for a program that performs the following calculations: $B \leftarrow A - 2$ (A and 2 are integers in two's complement format. The user types the value of A and the value of B is displayed on the monitor.)

ANS: 1FFE 240F 1040 B000 B000 2410 1041 2FF0 0000

or 10FE 240F 1040 B000 B000 2410 1041 2FF0 0000 (用不同 register)

八 (10 分) Using the instruction set of the simple computer as described in the class, write the code for a program that accepts two integers from the keyboard. If the first integer N1 is 0, the program increments the second integer N2; if the first integer is 1, the program decrements the second integer. The first integer must be only 0 or 1, otherwise the program fails. The program displays the result of the increment or the decrement on the monitor. (提示：可先輸入 0 到 Register0，再輸入題目所述的兩個數 N1 與 N2 後，比較 N1 是否 0 決定 N2 要加上 1 或減去 1)

ANS: 10FE 11FE 12FE D108 A200 2FF2 A100 D10A B200 2FF2 0000

(00) ₁₆	(10FE) ₁₆	// $R_F \leftarrow M_{FE}$, Input 0 from keyboard to R_0
(01) ₁₆	(11FE) ₁₆	// $R_F \leftarrow M_{FE}$, Input the first integer to R_1
(02) ₁₆	(12FE) ₁₆	// $R_F \leftarrow M_{FE}$, Input the second integer to R_2
(03) ₁₆	(D108) ₁₆	// If $R_0 \neq R_1$ then PC = 08, otherwise continue
(04) ₁₆	(A200) ₁₆	// $R_2 \leftarrow R_2 + 1$
(05) ₁₆	(2FF2) ₁₆	// $M_{FF} \leftarrow R_2$, Send the result to the monitor
(06) ₁₆	(A100) ₁₆	// $R_1 \leftarrow R_1 + 1$
(07) ₁₆	(D10A) ₁₆	// If $R_0 \neq R_1$ then PC = 0A, otherwise continue
(08) ₁₆	(B200) ₁₆	// $R_2 \leftarrow R_2 - 1$
(09) ₁₆	(2FF2) ₁₆	// $M_{FF} \leftarrow R_2$, Send the result to the monitor
(0A) ₁₆	(0000) ₁₆	// Halt